

# The empirical recurrence for OEIS A210407, proved by counting patterns

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31 August 2026

## Abstract

OEIS A210407 counts arrays subject to a local condition *and* to the clause “new values 0..3 introduced in row major order”, which says the array is written in canonical form. That clause is not local, so a transfer matrix over the alphabet does not see it. It does not have to: what is being counted is not arrays but equality patterns, and the number of patterns is recovered from the numbers  $L_i$  of arrays over an  $i$ -letter alphabet by inverting a falling factorial. The result is the clean identity  $4! a(n) = \sum_i \binom{4}{i} D_{4-i} L_i(n)$  with  $D_m$  the derangement numbers, and each  $L_i$  is an ordinary walk count. A second reduction — the chain is strongly lumpable over the patterns, because the condition is relabelling-invariant — brings the state space down from 100 to  $S = 15$ . The entry’s empirical recurrence of order 4, contributed by R. H. Hardin, then follows from a finite exact computation.

2020 Mathematics Subject Classification. 05A15, 05A18, 15B36.

## 1 The sequence and the conjecture

OEIS A210407 is “Number of  $(n+1) \times 3$  0..3 arrays with every  $2 \times 2$  subblock having three or four distinct values, and new values 0..3 introduced in row major order.”. It has offset 1 and begins

78, 2520, 80496, 2574592, 82334080, 2633043200, 84204544256, 2692856218368, ...

The entry carries the following comment:

Empirical:  $a(n) = 31*a(n-1) + 40*a(n-2) - 280*a(n-3) + 96*a(n-4)$ .

As of the “Last modified” line on the live entry (Jul 15 2018 12:03:46, revision 8) the statement is still recorded as empirical, and nothing on the entry records it as settled either way.

Written out, the claim is

$$a(n) = 31 a(n-1) + 40 a(n-2) - 280 a(n-3) + 96 a(n-4) \quad (1)$$

for all  $n > 4$ .

## 2 What the canonical-form clause counts

Fix the shape: 3 columns and  $L = n + 1$  rows, so  $L \cdot 3$  cells. Call a partition of the cells into nonempty classes a *pattern*; an array over an alphabet determines a pattern, two cells lying in the same class exactly when they carry the same value.

Every  $2 \times 2$  subblock of the array occupies two consecutive rows and two consecutive columns; writing those two rows as  $r$  and  $s$  the subblock is

$$\begin{pmatrix} \alpha & \beta \\ \gamma & \delta \end{pmatrix} = \begin{pmatrix} r_j & r_{j+1} \\ s_j & s_{j+1} \end{pmatrix},$$

and the entry's requirement on it is

$$\#\{\alpha, \beta, \gamma, \delta\} \in \{3, 4\}. \quad (2)$$

The condition is stated purely in terms of equalities between entries, so it is invariant under any relabelling of the values and is therefore a property of the pattern alone. Call a pattern *admissible* when it has the property.

The clause “new values 0..3 introduced in row major order” selects, from each class of arrays that differ only by a relabelling, exactly one representative: reading the cells in row major order, the first occurrence of the value  $v$  must precede the first occurrence of  $v + 1$ . Every pattern with at most  $K = 4$  classes has exactly one such representative, and every array in canonical form is the representative of its own pattern. Hence, writing  $N_j(n)$  for the number of admissible patterns with exactly  $j$  classes,

$$a(n) = \sum_{j=0}^4 N_j(n). \quad (3)$$

**Lemma 1.** *Let  $L_i(n)$  be the number of arrays over an alphabet of  $i$  letters satisfying the local condition, with no canonical-form clause. Then*

$$L_i(n) = \sum_j N_j(n) i(i-1) \cdots (i-j+1), \quad 4! \sum_{j=0}^4 N_j(n) = \sum_{i=0}^4 \binom{4}{i} D_{4-i} L_i(n),$$

where  $D_m = m! \sum_{t=0}^m (-1)^t / t!$  is the number of derangements of  $m$  letters.

*Proof.* An array over  $i$  letters satisfying the condition determines an admissible pattern together with an injection of its classes into the alphabet, and conversely; a pattern with  $j$  classes admits  $i(i-1) \cdots (i-j+1)$  injections. That is the first identity. Writing  $i(i-1) \cdots (i-j+1) = j! \binom{i}{j}$  it reads  $L_i = \sum_j (j! N_j) \binom{i}{j}$ , so by binomial inversion  $j! N_j = \sum_i (-1)^{j-i} \binom{j}{i} L_i$ . Summing over  $j \leq 4$  and exchanging the order of summation,

$$\sum_{j=0}^4 N_j = \sum_{i=0}^4 L_i \sum_{j=i}^4 \frac{(-1)^{j-i}}{i! (j-i)!} = \sum_{i=0}^4 \frac{L_i}{i!} \sum_{t=0}^{4-i} \frac{(-1)^t}{t!},$$

and multiplying by  $4!$  turns the coefficient of  $L_i$  into  $\frac{4!}{i!} \sum_{t \leq 4-i} (-1)^t / t! = \binom{4}{i} (4-i)! \sum_{t \leq 4-i} (-1)^t / t! = \binom{4}{i} D_{4-i}$ , an integer.  $\square$

### 3 Each $L_i$ is a walk count, and the chain lumps

Fix  $i$  and let  $V_i = \{0, \dots, i-1\}^3$  be the possible rows. For  $r, s \in V_i$  declare  $r \rightarrow s$  an edge when placing  $s$  directly after  $r$  satisfies (2) at every column, and let  $M_i$  be the adjacency matrix. An array is exactly a sequence of rows, and its condition is exactly the conjunction of the edge conditions along that sequence, so

$$L_i(n) = \mathbf{s}_i^\top M_i^n \mathbf{1}.$$

Because the condition involves no row before the first, every row may start a walk, and  $\mathbf{s}_i = \mathbf{1}$ .

**Lemma 2.** Let  $\pi(r)$  be the equality pattern of the row  $r$ . For all  $r, r'$  with  $\pi(r) = \pi(r')$  and every pattern  $Q$ ,

$$\#\{s : r \rightarrow s, \pi(s) = Q\} = \#\{s : r' \rightarrow s, \pi(s) = Q\}.$$

Consequently the numbers  $\widehat{M}_i[P][Q]$  obtained by taking any representative of  $P$  are well defined, and with  $e_P = i(i-1)\cdots(i-|P|+1)$  the number of rows of pattern  $P$ ,

$$\mathbf{s}_i^\top M_i^m \mathbf{1} = \sum_P \varepsilon_P f_m(P), \quad f_0 \equiv 1, \quad f_{m+1}(P) = \sum_Q \widehat{M}_i[P][Q] f_m(Q),$$

where  $\varepsilon_P = e_P$  if rows of pattern  $P$  may start a walk and 0 otherwise.

*Proof.* If  $\pi(r) = \pi(r')$  then the map sending the value of  $r$  at a column to the value of  $r'$  at that column is a well-defined injection between the letters used, and extends to a permutation  $\sigma$  of the alphabet with  $r' = \sigma(r)$ . The edge condition is a statement about equalities among entries, so  $r \rightarrow s$  if and only if  $\sigma(r) \rightarrow \sigma(s)$ ; and  $\pi(\sigma(s)) = \pi(s)$ . Hence  $s \mapsto \sigma(s)$  is a bijection between the two successor sets preserving patterns, which is the first claim. It says the chain is strongly lumpable over  $\pi$ , so  $(M_i^m \mathbf{1})(r)$  depends on  $r$  only through  $\pi(r)$ ; calling that value  $f_m(\pi(r))$  gives the recursion, and summing over  $r$  in each class gives the displayed identity, the number of rows of pattern  $P$  being the number of injections of its  $|P|$  classes into the alphabet.  $\square$

Assembling the 4 lumped chains into one block-diagonal matrix  $N$  of size  $S = 15$  and putting the weights of Lemma 1 into the start vector,  $\mathbf{v} = ((\binom{4}{i} D_{4-i} \varepsilon_P)_{i,P})$ , gives

$$4! a(n) = \mathbf{v}^\top N^n \mathbf{1}. \quad (4)$$

The unlumped alphabet had 100 rows in all; the lumped chain has  $S = 15$  states.

## 4 The criterion, and the computation

Let  $q(t) = t^4 - \sum_i c_i t^{4-i}$  be the characteristic polynomial of (1).

**Lemma 3.** For every  $n$  with  $m = n + 1 - 1 \geq 4$ ,

$$4! \left( a(n) - \sum_i c_i a(n-i) \right) = \mathbf{v}^\top N^{m-4} q(N) \mathbf{1}.$$

Hence (1) holds from that point on if and only if  $\mathbf{v}^\top N^j q(N) \mathbf{1} = 0$  for every  $j \geq 0$ , and it is enough to check  $j = 0, 1, \dots, S-1$ .

*Proof.* Substitute (4) into the left side and factor  $N^{m-4}$  out of  $N^m - \sum_i c_i N^{m-i}$ . The scalar  $4!$  is nonzero, so it does not affect whether the left side vanishes. For the last clause put  $w = q(N) \mathbf{1}$ : by Cayley–Hamilton  $N^S$  is an integer combination of  $I, N, \dots, N^{S-1}$ , so each  $\mathbf{v}^\top N^j w$  with  $j \geq S$  is the corresponding combination of earlier values.  $\square$

**Theorem 4.** The recurrence (1) holds for every  $n > 4$ , which is the statement of Section 1.

*Proof.* The lumped transition counts were obtained by enumerating, for one representative row of each pattern, all admissible successors and sorting them by pattern — the successors being generated column by column, since the edge condition is a conjunction of one constraint per column. The vector  $w = q(N) \mathbf{1}$  was then formed by 4 matrix–vector products in exact integer arithmetic and  $\mathbf{v}^\top N^j w$  evaluated for  $j = 0, 1, \dots$ , again exactly. Every one of those integers vanishes from the index corresponding to  $n = 4 + 1$  onwards, and the run continued until  $w$  was the zero vector, which settles all larger  $j$  at once. Lemma 3 gives the theorem.  $\square$

## 5 Verification

Three checks, each independent of the algebra above.

First, the whole construction was compared against the entry's DATA: the right-hand side of (4), divided by  $4!$ , was evaluated at every one of the 15 published indices and agrees exactly. This is a strong test of the modelling and not only of the arithmetic — the inversion of Lemma 1, the reading of the local condition, and the alignment of the entry's index with the number of rows would each show up as a mismatch, and the quotient by  $4!$  came out an integer at every index.

Second, the lumped chain was checked against the unlumped one: for the small shapes the walk counts  $\mathbf{s}_i^\top M_i^m \mathbf{1}$  were computed directly over the full alphabet as well and agree with  $\sum_P \varepsilon_P f_m(P)$  term by term.

Third, the recurrence (1) was evaluated on the published terms in exact integer arithmetic with no matrices involved, and the annihilation test of Lemma 3 was run until the working vector was identically zero rather than to a sampled prefix.

## References

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